

## NEWTON'S LAW OF COOLING (07)

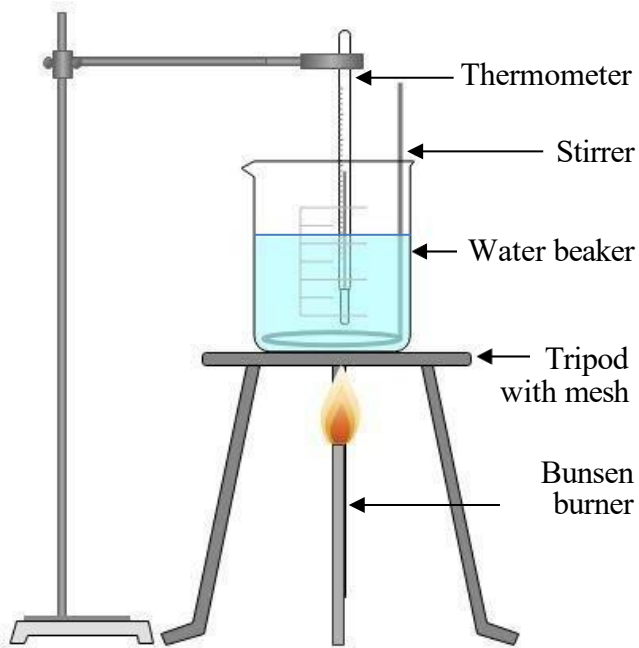
### Aim

To verify Newton's law of cooling.

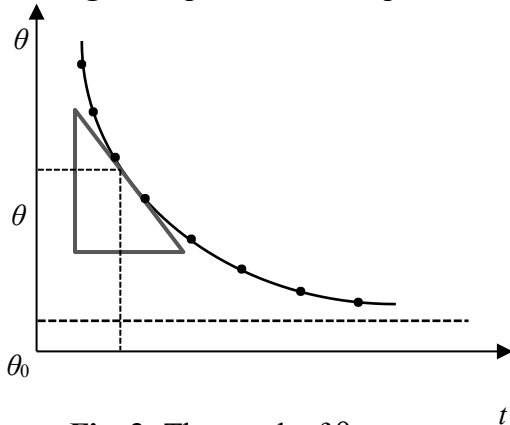
### Materials and Apparatus

Tripod with mesh, water beaker, Bunsen burner, calorimeter with stirrer and lid, two metal containers with different volumes, two thermometers (0-100 °C), stirrer, and stopwatch.

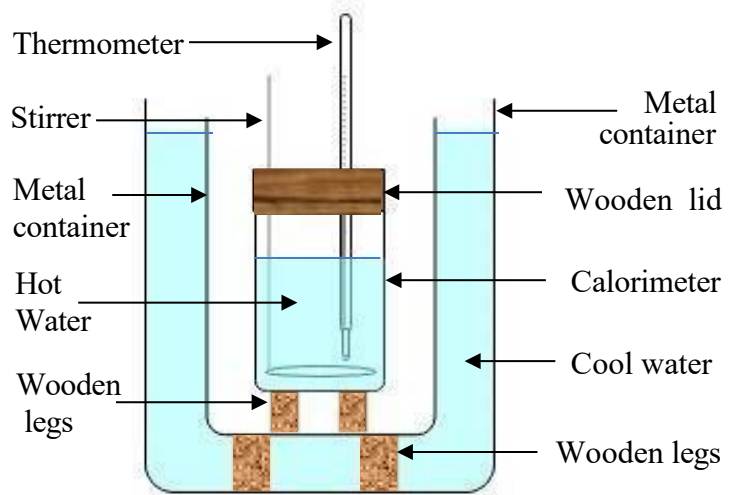
### Theory



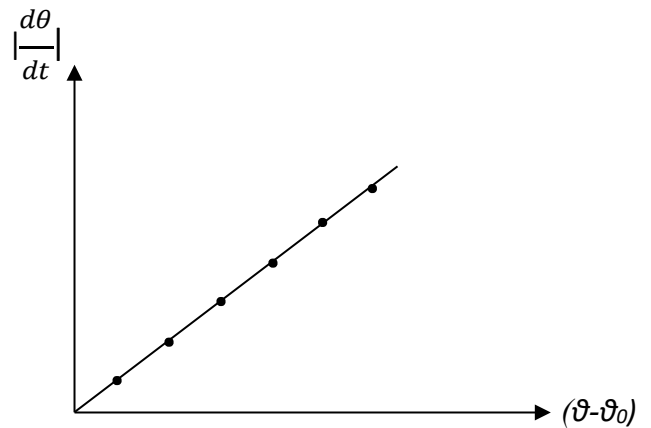
**Fig. 1:** Experimental Setup



**Fig. 3:** The graph of  $\theta$  vs  $t$



**Fig. 2:** Experimental setup



**Fig. 4:** The graph of  $|d\theta/dt|$  vs  $(\theta - \theta_0)$

**NEWTON'S LAW OF COOLING (07)**

According to Newton's cooling law, the rate of loss of heat from a hot body due to convection can be written as,

$$\frac{dQ}{dt} = \varepsilon A(\theta - \theta_o) \quad (1)$$

where,

$\frac{dQ}{dt}$  : Heat loss rate

$A$  : Area of calorimeter

$\theta$  : Temperature of the hot body

$\theta_o$  : Temperature of surroundings

$\varepsilon$  : Emissivity of the calorimeter

$$\frac{dQ}{dt} = (m_c s_c + m_w s_w) \frac{d\theta}{dt} \quad (2)$$

where,

$m_c$  : Mass of the calorimeter

$s_c$  : Specific heat capacity of the calorimeter material

$s_w$  : Specific heat capacity of water

$m_w$  : Mass of water

From equations (1) and (2), obtain a relationship between  $\frac{d\theta}{dt}$  vs  $(\theta - \theta_o)$

**Procedure**

- a) Set up the apparatus as shown in Fig. 1.
- b) Heat the water to about 55 °C.
- c) Set up the apparatus as shown in Fig. 2.
- Q.** What are the precautions to be taken in setting up the apparatuses in Fig.1 and Fig. 2
- d) Fill the calorimeter with the heated water up to about 3/4 of the volume.
- e) Start writing down the temperature every 30 seconds until the temperature drops below 35 °C (to obtain a good cooling curve).
- f) Note the room temperature  $\theta_o$ .
- g) Plot the temperature  $\theta$  versus time  $t$ .
- h) Obtain  $d\theta/dt$  values at 6 different temperatures from the graph that you plot.
- i) Plot the graph  $\frac{d\theta}{dt}$  vs  $(\theta - \theta_o)$  to verify Newton's law of cooling.
- Q.** What sort of a graph is expected here?